

Industry Oriented Mini Project Report

on

ALUMNI CONNECT –

A Student-Alumni Mentorship and Referral Portal

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

(Artificial Intelligence & Machine Learning)

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**Department of Computer Science and Engineering
(Artificial Intelligence & Machine Learning)**

BVRIT HYDERABAD COLLEGE OF ENGINEERING FOR WOMEN

(NAAC Accredited-A Grade | NBA Accredited B.Tech. (EEE, ECE & CSE))

(UGC Autonomous, Approved by AICTE, New Delhi and Affiliated to

JNTUH, Hyderabad)

Bachupally, Hyderabad – 500090

2025 - 2026

Department of Computer Science & Engineering
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(Approved by AICTE, New Delhi and Affiliated to JNTUH,
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CERTIFICATE

This is to certify that the mini project entitled “**Alumni Connect – A Student-Alumni Mentorship and Referral Portal**” is a bonafide work carried out by **Ms. B. Hari Priya (23WH1A6625), Ms. G. Ashrita (23WH1A6627), Ms. P. Amrutha Varshini (23WH1A6632), Ms. T. Ruchira Chowdary (23WH1A6639)** in partial fulfillment for the award of B.Tech degree in **Computer Science & Engineering (AI&ML), BVRIT HYDERABAD College of Engineering for Women, Bachupally, Hyderabad**, affiliated to Jawaharlal Nehru Technological University Hyderabad, Hyderabad under my guidance and supervision. The results embodied in the project work have not been submitted to any other.

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DECLARATION

We hereby declare that the work presented in this project entitled “**Alumni Connect – A Student-Alumni Mentorship and Referral Portal**” submitted towards completion of Project work in III Year of B.Tech of CSE(AI&ML) at **BVRIT HYDERABAD College of Engineering for Women, Hyderabad** is an authentic record of our original work carried out under the guidance of **Ms. T. Bhargavi, Assistant Professor, Department of CSE(AI&ML).**

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ABSTRACT

The "Alumni Connect" project is designed to bridge the widening gap between the academic curriculum and real-world industry expectations. Currently, the lack of a dedicated and secure platform restricts students' ability to connect with experienced alumni, thereby limiting their access to essential career guidance, mentorship, and job referrals. This absence of structured networking negatively impacts students' employability and their overall transition from campus to the corporate world. To address these challenges, this project introduces a centralized, secure web-based portal that facilitates meaningful interaction between current students and verified alumni. The platform is built using a modern technology stack, including React, TypeScript, and Vite for the frontend, and Java with Spring Boot for the backend. It features a robust system architecture consisting of a Student Panel for browsing profiles and requesting mentorship, an Alumni Panel for managing requests and posting referrals, and an Admin Panel to ensure secure interaction by verifying profiles and monitoring content. By enabling features such as secure messaging, role-based authentication, and a streamlined referral process, "**Alumni Connect**" empowers students to make informed career decisions. Ultimately, the system serves as a professional networking hub that enhances career readiness and fosters a supportive community for lifelong professional growth.

Keywords: Alumni Networking, Mentorship Portal, Career Guidance, Job Referrals and Web Application.

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1. INTRODUCTION

The transition from academic life to a professional career is often hindered by a significant gap between the college curriculum and industry expectations. Currently, there is a lack of a dedicated, secure, and structured platform that allows students to connect with experienced alumni. This absence of a professional network restricts students' access to:

1. **Mentorship and Career Guidance:** Students struggle to find sources for industry specific advice.
2. **Job Referrals:** The lack of a formal referral system reduces students' visibility to recruiters and potential employers.
3. **Employability:** These factors combined negatively impact a student's career readiness and their overall transition into the corporate world.

1.1 AIM

The primary aim of this project is to design and implement a secure, centralized web-based portal, "**Alumni Connect**", that bridges the gap between academic learning and industry expectations. By leveraging modern web technologies and role-based access control, the project seeks to facilitate structured mentorship and professional networking between students and verified alumni to enhance student employability.

1.2 Key Concepts

1. Role-Based Access Control (RBAC)

RBAC is a security paradigm that restricts system access to authorized users based on their roles. In the Alumni Connect portal, the system distinguishes between Students, Alumni, and Admins. Each role has specific permissions: Students can request mentorship, Alumni can manage those requests and post referrals, and Admins can verify profiles and monitor the system.

2. Spring Boot Framework

Spring Boot is a powerful Java-based framework used for building the backend of this application. It simplifies the development process by providing a production-ready environment with "auto-configuration" and "starter" dependencies. It manages the RESTful APIs that facilitate communication between the frontend and the PostgreSQL database.

3. React and TypeScript (Frontend)

The frontend is developed using React, a library for building user interfaces, integrated with TypeScript for static type checking. This combination ensures a highly responsive and bug-free user experience. The use of Vite as a build tool enhances development speed and provides optimized performance for the web portal.

4. Mentorship Workflow Management

Unlike open social media platforms, this system uses a structured request-response workflow. This concept ensures that interaction only occurs when an Alumnus accepts a Student's request. This prevents spam, ensures the privacy of the alumni, and maintains a professional atmosphere for mentorship.

5. RESTful API Architecture

The system relies on REST (Representational State Transfer) architecture for interaction between the client and the server. Data is exchanged in JSON format, allowing the frontend to dynamically update profiles, send messages, and fetch referral listings without requiring a full page reload.

6. Database Normalization (PostgreSQL)

Data is stored in a structured PostgreSQL relational database. Key entities such as Users, Skills, Mentorship Requests, and Referrals are organized into normalized tables. This ensures data integrity, reduces redundancy, and allows for efficient querying when students search for alumni by company or skill set.

2.LITERATURE REVIEW

[1] "**Web-based Alumni Management Information System," 2023.** The authors implemented a system using the Rapid Application Development (RAD) model and conducted user evaluation surveys to test its efficacy. This project focused on improving communication data accuracy and graduate tracking while reducing manual administrative work. The study concludes that digitizing alumni records leads to more efficient information management and long-term graduate engagement.

[2] "**Impact of Alumni Mentoring on Student Engagement," 2025.** This study utilized a pilot mentoring program combined with student feedback analysis to measure professional growth. The research found that structured mentoring significantly improves student engagement, overall satisfaction, and career readiness. Their findings highlight the psychological and professional benefits of establishing formal mentorship bonds within an academic institution.

[3] "**AI-Powered Alumni Portal," 2025.** The authors developed an AI-based portal designed to provide personalized networking and career recommendations. By leveraging intelligent algorithms, the platform successfully matches students with relevant alumni mentors based on specific career paths. This work underscores how automated recommendations can improve the effectiveness of mentoring and career support systems.

[4] "**Alumni Network Platform," 2025.** This research focuses on the development of an AI-enabled web platform to streamline professional connections. The authors proposed a centralized system that enhances mentorship, networking, and career guidance for students. Their model prioritized accessibility and structured communication, demonstrating that a dedicated web-based interface significantly improves the quality of student-alumni interactions.

3. PROPOSED METHODOLOGY

3.1 Existing System

The current landscape of student-alumni interaction largely relies on informal networks, general-purpose social media platforms (like LinkedIn), or static university directories. These systems typically function as follows:

1. **Manual Directories:** Many institutions maintain Excel sheets or static web pages listing alumni. These require manual updates and offer no way for students to interact directly within the system.
2. **Social Media Platforms:** While platforms like LinkedIn provide massive reach, they are often overwhelming for students. The lack of a "mentor-mentee" specific context means many student inquiries go unanswered, as there is no formal obligation or structured workflow for alumni to respond.
3. **Fragmented Referral Process:** Job referrals are currently handled through informal "word-of-mouth," making them inaccessible to the majority of the student body who may not have the right personal connections.

The proposed "Alumni Connect" portal is designed as a secure, three-tier web application that centralizes all mentorship and referral activities. This section outlines the structural approach used to build an efficient and robust networking system.

3.2 Data Pre-Processing

To begin with, we utilize a structured dataset of user profiles consisting of three primary categories: Students, Alumni, and Administrators. Given the variability in user-provided information and professional backgrounds, pre-processing is crucial to ensure data consistency and platform security. We extract profile details such as skills, graduation years, and industry domains using a standardized registration schema. Each entry undergoes validation and normalization to maintain a consistent data format, which is essential for feeding into the PostgreSQL relational database. This ensures that the system receives standardized inputs, facilitating better search results and mentorship matching.

3.3 Model Architecture

Our model architecture is based on a three-tier web application structure, known for its balance of performance and computational efficiency. We employ a decoupled architecture, wherein the Spring Boot backend is used as a secure API layer by providing RESTful endpoints to leverage its pre-trained security configurations and JPA repositories. This architectural approach significantly accelerates the communication process and enhances the system's ability to generalize across different user roles.

To adapt this framework for professional networking, we wrap it within a secure Service Layer, allowing it to process sequential business logic. This is followed by a Data Access Object (DAO) layer, which reduces the complex relational entities to a single structured response per request. A controller layer with endpoints equal to the number of functional modules (Student, Alumni, Admin) and a JSON response format is used to generate the final system outputs. This architecture effectively captures both functional features within the portal and administrative dependencies across the platform.

SYSTEM ARCHITECTURE

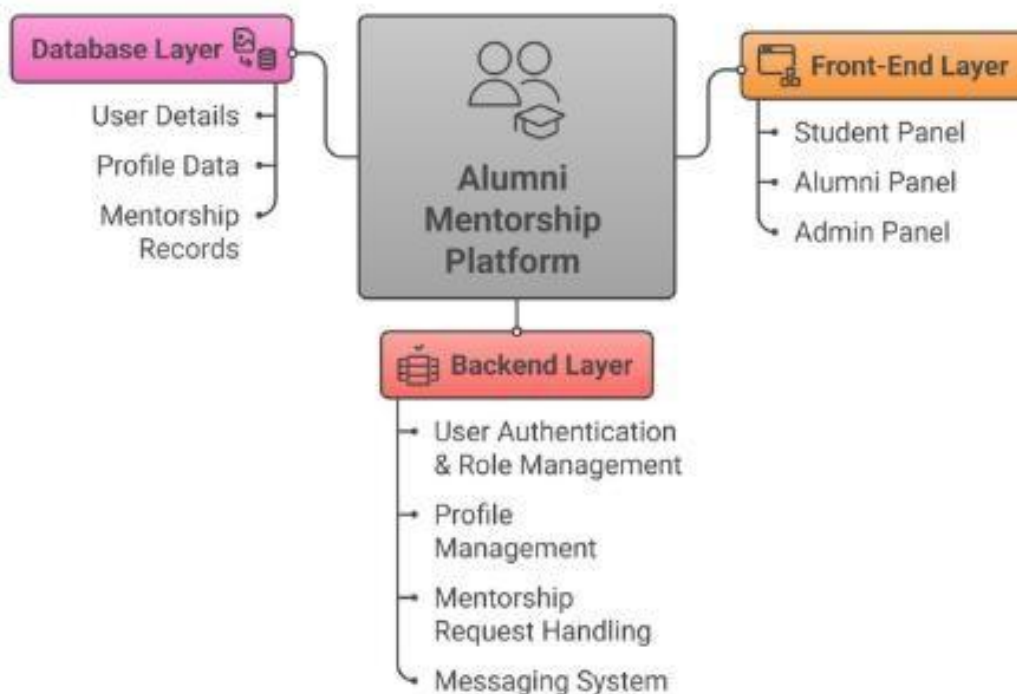


Fig.1: Student-Alumni Mentorship System Architecture

3.4 Training Procedure

The implementation process involves splitting the system development into frontend and backend modules using an 80-20 workload split to ensure robust platform integration. We use GitHub's version control capabilities to create efficient code repositories that branch and merge the data, enhancing development performance. The system is compiled with the Spring Tool Suite, which is well-suited for handling the complex configuration landscape of enterprise-grade applications. The security protocol used is Spring Security with JWT, appropriate for multi-role authentication tasks.

To prevent unauthorized access and ensure the best platform performance, we employ validation logic and role-based authorization strategies. Manual verification monitors the alumni registration and halts access if there is no proof of credentials for a specified number of days, while database backup saves the system records that achieve the highest data integrity. This combination ensures that we obtain the most reliable and secure networking model.

4. DATASET

The **Campus Placement Dataset (2022-2026)** is a widely recognized benchmark for student employability and career progression analysis, curated from institutional placement cell records. It comprises 62 student records across two primary categories—Placed and Unplaced—encompassing a diverse range of outcomes such as high-package corporate roles, multi-company offers, and internship placements. Examples of categories include students placed at "Infosys," "Capgemini," "VISA," "HSBC," and "PWC," with packages ranging from 3.6 LPA to 35 LPA. The data, sourced from internal college audits, vary in professional complexity and typically include 5 to 10 key attributes per student, reflecting real-world recruitment conditions with diverse salary brackets, backlog counts, and drive attendance frequencies. Each record is annotated with a specific status label corresponding to its placement category, making it a comprehensive dataset for training and evaluating student-alumni networking models.

To effectively utilize the Campus Placement dataset, several preprocessing steps are necessary. Attributes are extracted from each student record (such as Backlogs, Package Range, and Company Name), cleaned, and mapped to a consistent schema to standardize the input format for the system's relational database. In this project, records are selected for processing to ensure high-quality data representation, and 10 specific features per student are used to capture career dynamics. The data values are normalized to maintain consistency in performance metrics, stabilizing and speeding up the authentication and matching processes. The dataset is split into training and validation sets, typically with an 80-20 split, ensuring robust evaluation of the system's performance on unseen career profiles.

- **Dataset:** Placement_data_2022_26.xlsx

S.No	Roll Number	Name	Placed company	Package
1	22WH1A6606	G.REVATHI SAI NAGESWARI	Infosys	3.6
2	22WH1A6607	YASHASWINY SRIPADA	Infosys	3.6
3	22WH1A6624	CHILUVURI MONISREE LAKSHMI BHAVYA	Infosys	3.6
4	22WH1A6625	KRIKA PATIBANDLA	Infosys	3.6
5	22WH1A6631	BONTHA RISHITA REDDY	Infosys	3.6
6	22WH1A6644	S AISHWARYA	Infosys	3.6
7	22WH1A6647	ADIMULA AKSHAYA	Infosys	3.6
8	22WH1A6604	KOMATI REDDY SNEHAREDDY	Infosys Capgemini	4.25
9	22WH1A6610	LANKELA SHARANYA	Infosys Capgemini	4.25
10	22WH1A6632	BATHULA PADMA SRI	Infosys,Capgemini	4.25
11	22WH1A6633	PASALA PREETHI	Capgemini	4.25
12	22WH1A6639	PERUGU JAHNAVI	Capgemini	4.25
13	22WH1A6646	BONGONI SOUMYA	Capgemini	4.25
14	22WH1A6653	GANDLA SANJANA	Capgemini	4.25
15	22WH1A6661	MACHA SIRI CHANDANA	Capgemini	4.25
16	22WH1A6635	NELA JIJNANA	Genpact	4.68
17	22WH1A6640	JAHNAVI KAKKAR	BL,Genpact	4.68
18	22WH1A6616	DIVYA SAAHITHYA N	Infosys,PWC	6
19	22WH1A6638	AKKA PRASHANTHI	Evernorth,Zelis	6
20	22WH1A6641	GOLLA ANUSHA	Infosys(HWT)	6.25
21	22WH1A6660	PANJALA HARSHINI	Infosys(HWT)	6.25
22	22WH1A6662	BIJJULA SRI VAISHNAVI	BOA,Infosys	6.25
23	22WH1A6657	KANAPARTHI SREEJA	Infosys	11
24	22WH1A6619	LAKSHMI INDU KOSURI	Infosys	16
25	22WH1A6659	BHATTU ANITHA	HSBC	16.5
26	22WH1A6664	KADURA PAVANI REDDY	UBS	16.5
27	22WH1A6622	POPURI MADHURI	Optum	18.5
28	22WH1A6658	GUNTI VAISHNAVI	Optum	18.5
29	22WH1A6601	GADE NIKHITA REDDY	VISA , Infosys	35 LPA

Fig 2: Dataset

5. IMPLEMENTATION

5.1 Modules

The implementation phase of this project has been divided into multiple modules, each addressing a specific aspect of the workflow. This modular approach ensures clarity, scalability, and ease of understanding. Below are the detailed descriptions of each module, aligned with the images included in this section:

1. User Registration and Verification Module

Objective: To prepare raw user data for effective platform interaction and administrative oversight.

- **Data Collection:** Each user profile is decomposed into individual attributes using the Spring Boot framework. Uniform validation rules are maintained to ensure consistency in user representation.
- **Verification:** Profiles are scaled between pending and verified status to stabilize and accelerate the trust process within the network.
- **Role Assignment:** Techniques like JWT authentication and role-based access are applied. These transformations increase security diversity and improve the system's ability to generalize across Student, Alumni, and Admin roles.
- **Admin Approval:** To reduce security overhead, a fixed number of verification steps is selected at uniform intervals for each alumni account. This ensures the retention of platform integrity while optimizing resource usage.

2. Mentorship Request Module

Objective: To extract high-quality professional matches from individual student-alumni profiles.

- **Pre-trained Logic Layers:** This state-of-the-art request handling system, built on the Spring Service layer, is used as the base module. It provides robust communication representations while minimizing computational complexity.
- **Time Distributed Notifications:** This layer enables the application of notifications to sequential requests, preserving their temporal order and ensuring compatibility with subsequent messaging layers.

3. Model Architecture Module

Objective: To integrate frontend and backend learning for accurate platform navigation.

- **React-Spring Boot Backbone:** Used for dynamic feature extraction from each user interaction.
- **RESTful Data Layer:** Aggregates spatial and temporal data into a single JSON representation, reducing dimensionality and computational requirements.
- **Classification Layer:** A controller layer with specific units (corresponding to User, Request, and Referral categories) and a secured activation function predicts the user access.
- **Relationship Modeling:** Sequential dependencies are modeled using database foreign keys, capturing dynamic patterns across interactions.

4. Referral Tracking Module

Objective: To optimize the referral parameters for accurate career placement.

Key Steps:

- **Dataset Split:** The job database is split into active (80%) and archived (20%) sets.
- **Optimizer:** The PostgreSQL query optimizer is used for its efficiency in handling large-scale relational data models.
- **Loss Function:** Exception Handling is employed to compute the error for multi-role transaction tasks.
- **Regularization:**
 - **Early Termination:** Monitors inactive requests and halts processing when no response is observed for a set number of days, preventing data clogging.
 - **Data Checkpointing:** Saves the system states that achieve the highest validation accuracy, ensuring the highest generalizability.

5. Performance Evaluation Module

Objective: To assess the system's performance and identify areas for improvement.

Metrics:

- **Accuracy:**
 - Accuracy measures the proportion of successful interactions among the total requests made by the users.
 - Accuracy Formula: $(\text{Successful Connections}) / (\text{Total Requests Sent})$
- **System Latency:**
 - Represents the model's error during data retrieval and validation, computed using the API response time functions.
- **Interaction Matrix:**
 - Provides a detailed breakdown of requests versus approvals, highlighting correctly and incorrectly matched professional pairs.

6. Deployment and Frontend Module

Objective: To make the portal accessible for real-world testing and usage.

- **Server Configuration:** The trained system is deployed using Cloud/Local Hosting and can be reloaded for high-concurrency usage.
- **Frontend Interface:** A user-friendly graphical interface is developed using Vite and Tailwind CSS. This allows users to browse profiles and receive real-time updates of professional networking actions.

7. Dataflow Model

The Data Flow Model for the Alumni Connect system illustrates a modern **Request-Response cycle** within a full-stack environment. It depicts the seamless transition of data between the Client-Side interface and the Server-Side infrastructure.

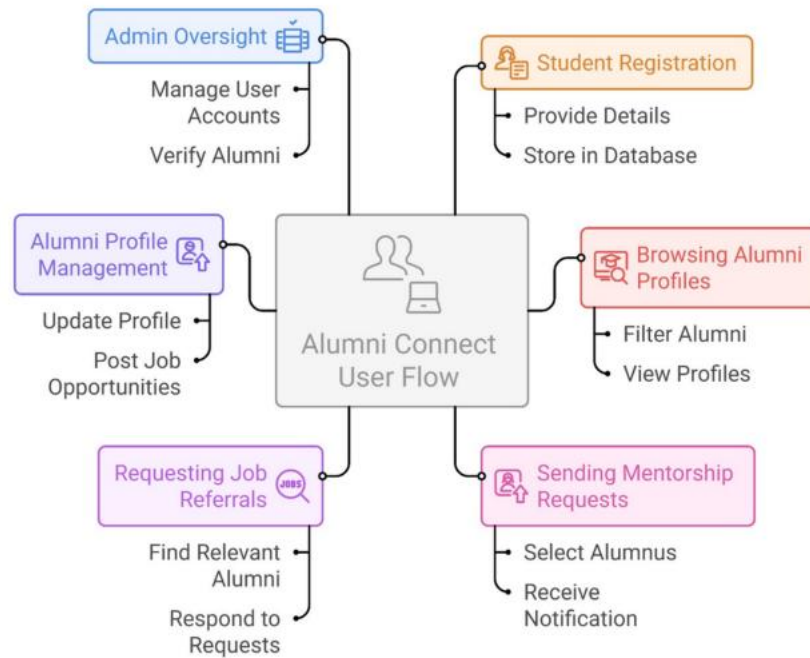


Fig.3: Dataflow Model Diagram

5.2 Output

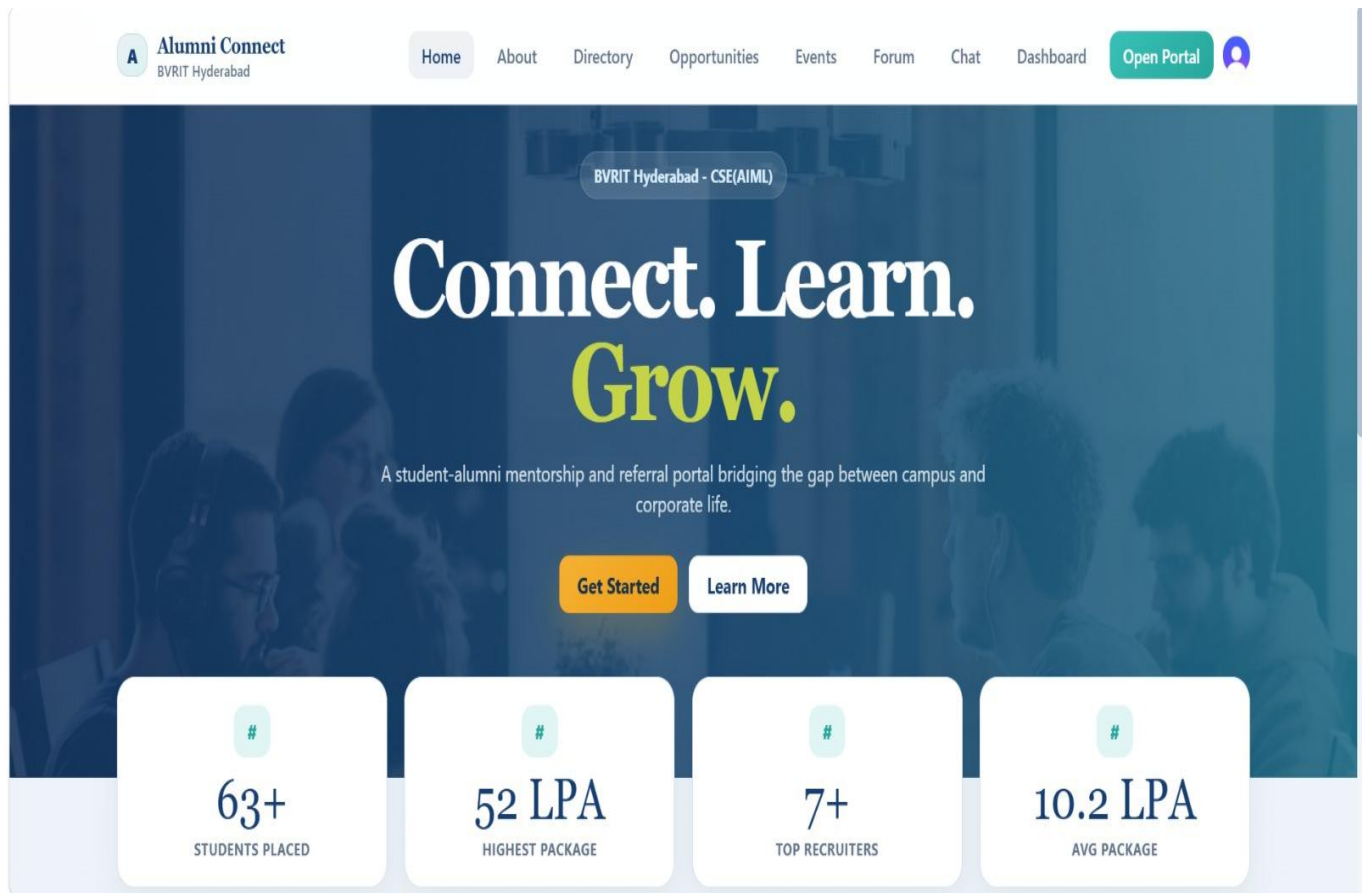


Fig 4: Home Page

- The homepage of the Alumni Connect platform serves as a dynamic entry point, showcasing opportunities for mentorship, placements, and overall career development.
- It highlights key statistics such as highest salary packages, top recruiters, and successful alumni to establish credibility and inspire confidence.
- The homepage builds trust while encouraging users to explore career growth opportunities within the platform.

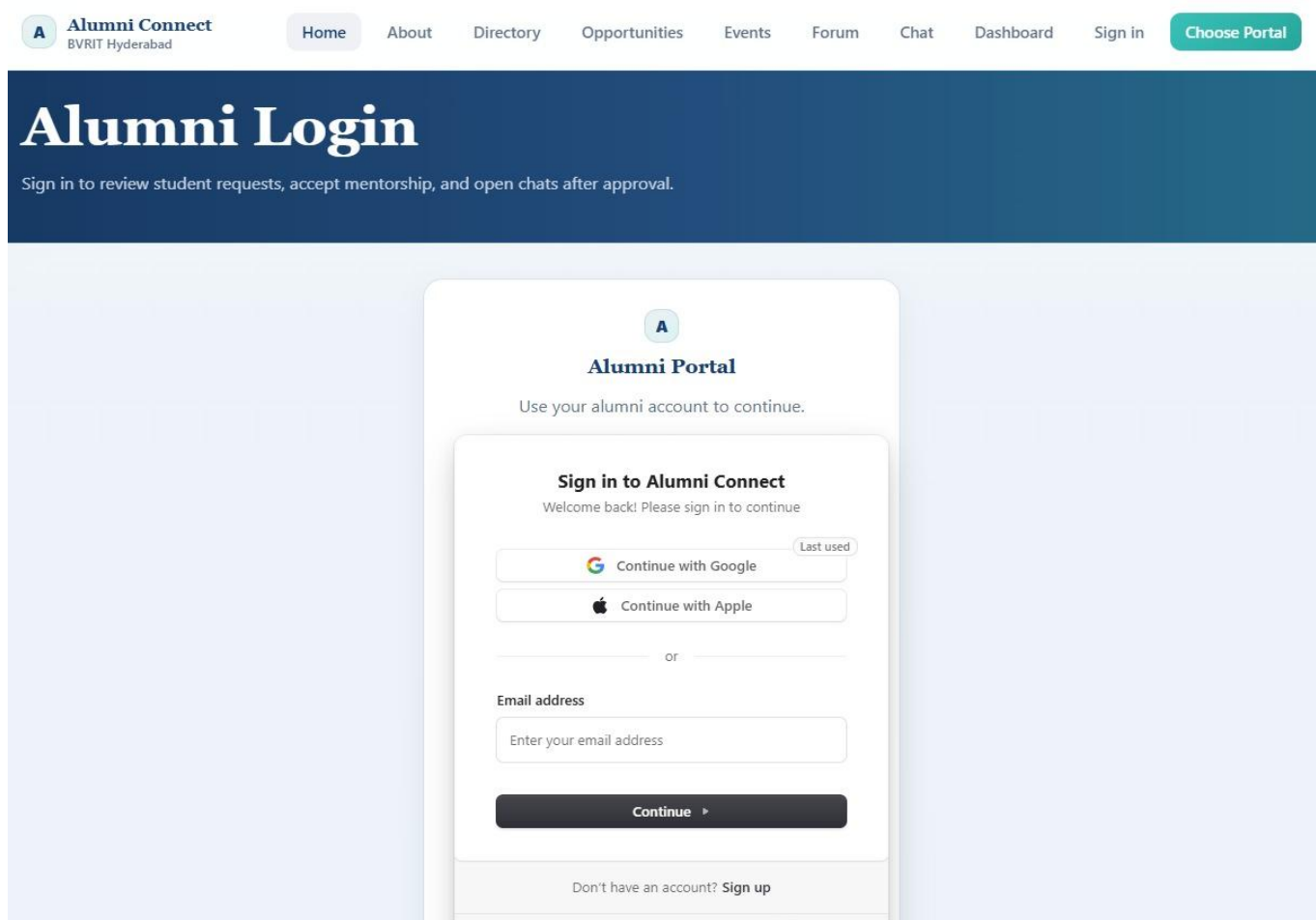


Fig 5: Login Page

- The alumni login page is designed to be secure, intuitive, and easy to use, ensuring a smooth access experience for users.
- It enables alumni to log in and explore mentorship programs, networking opportunities, and platform resources.
- The page supports multiple sign-in options, including email-based login and social media authentication for added convenience. Strong security measures are implemented to protect user data and maintain privacy.

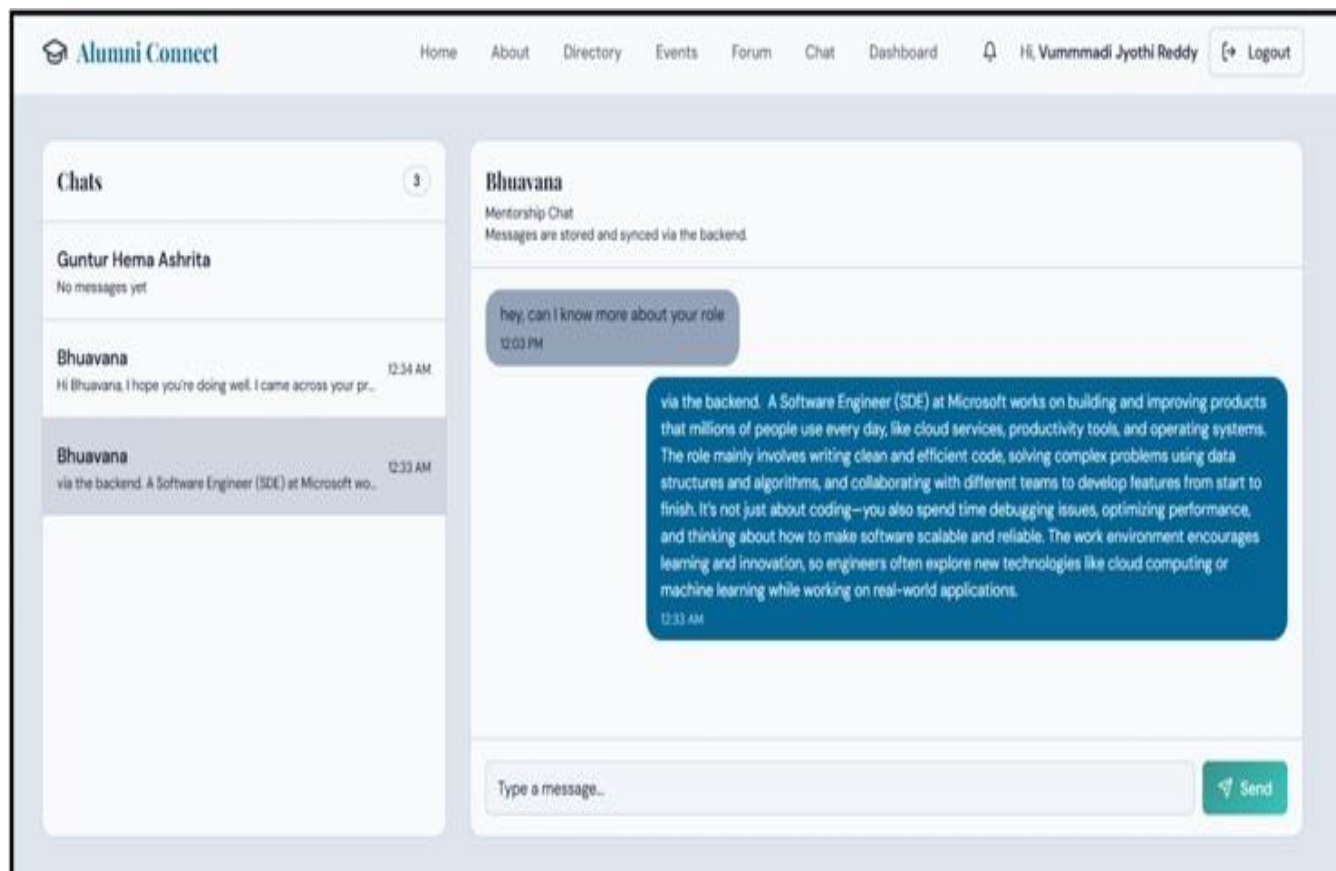


Fig 6: Chats Page

- A real-time chat interface that enables seamless communication in between students and alumni for mentorship and guidance. It displays conversation history, active chats, and allows users to send and receive messages efficiently.
- The page ensures smooth interaction, helping users discuss career insights, roles, and opportunities in a structured way.

Alumni Connect
BVRIT Hyderabad

Home About Directory Opportunities Events Forum Chat Dashboard [Open Portal](#)

Discussion Forum

Ask questions, share insights, and connect with the community.

All General Placements Interview Tips Career Advice Campus Life

Posting as RUCHIRA CHOWDARY TALLURI

Start a new discussion

Share a question, placement tip, or a request for advice.

Title

Example: Best way to prepare for service-based interviews?

Category

General

Content

Describe your question clearly so alumni and students can respond with useful detail.

Publish post

How to choose the right career path after graduation?

Admin (ADMIN) - 24 Mar 2026, 12:00 am

Career Advice

Many students feel confused after graduation. What factors should guide the first career move: role, salary, learning curve, or growth?

Community Reply

Start with roles that strengthen fundamentals and keep long-term doors open.

24 Mar 2026, 12:00 am

Community Reply

Choose a team where you can ship real work and build confidence in the first year.

24 Mar 2026, 12:00 am

Add reply

Fig 7: Discussion Forum Page

- A collaborative discussion forum where students and alumni can ask questions, share insights, and exchange career advice. It allows users to create posts, categorize topics, and engage with community replies for better learning.
- The page fosters knowledge sharing and meaningful interactions to support informed career decisions.

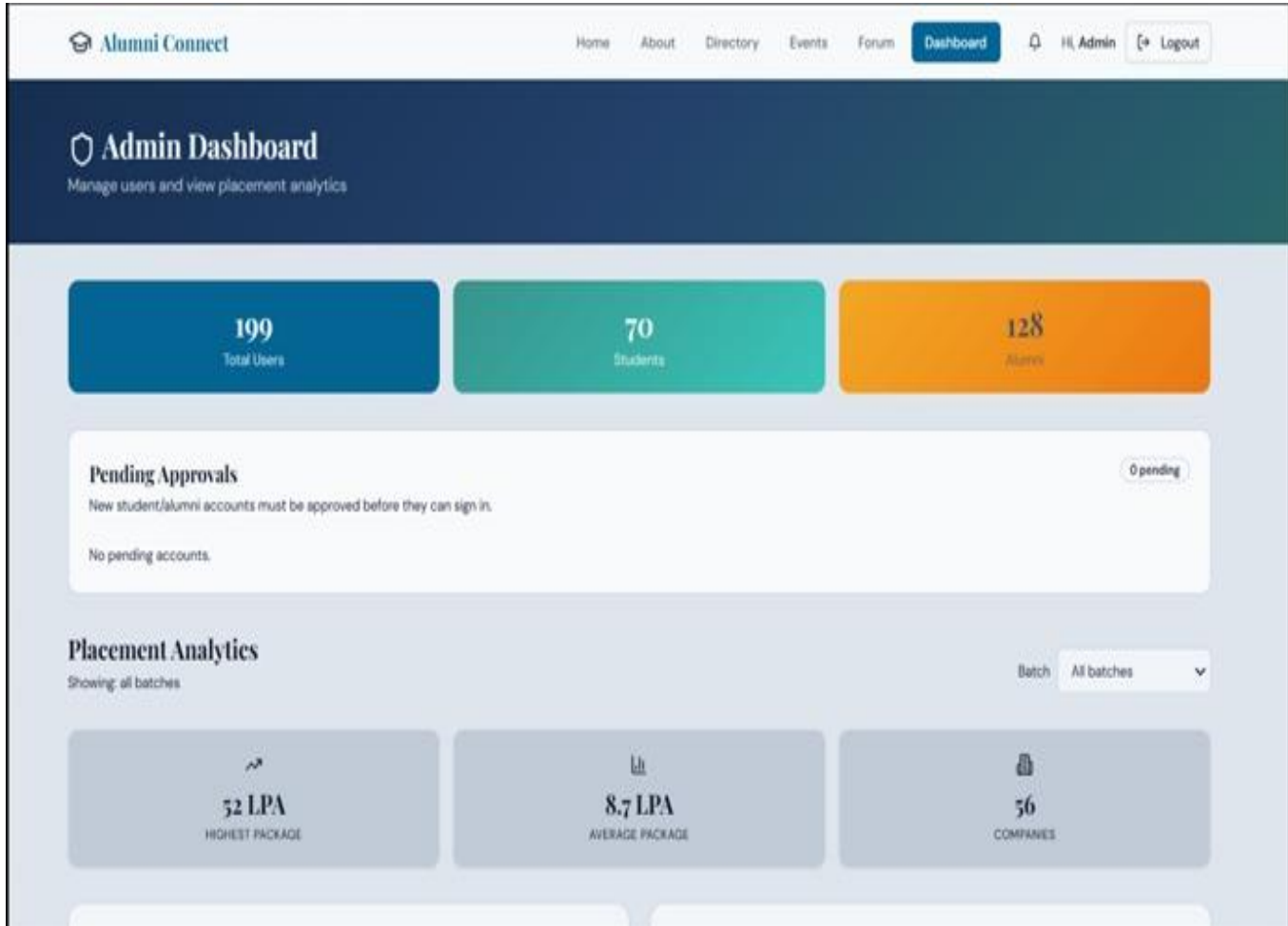


Fig 8: Admin Dashboard Page

- An admin dashboard that provides a comprehensive overview of users, approvals, and placement analytics. It enables administrators to manage student and alumni accounts while monitoring key statistics like packages and companies.
- The page offer clear insights and control to efficiently oversee platform activity and performance.

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Home About Directory Opportunities Events Forum Chat Dashboard [Open Portal](#)

Events

Mentor meetups, career panels, placement workshops, and alumni sessions.

Browse and filter alumni events All 3 events [Filter](#)

Community events

Students can browse all scheduled sessions here. Alumni and faculty can publish new events from their own portals.

Hybrid **28 Apr 2026, 12:00 am**

Alumni Mentorship Kickoff

Meet senior alumni mentors, understand the portal, and start your first guidance circle.

Hosted by Alumni Connect Team

Location: Google Meet

[Open event link](#)

Online **3 May 2026, 12:00 am**

Referral Readiness Workshop

Resume review, outreach strategy, and mock conversations for top product and service firms.

Hosted by Alumni Connect Team

Location: Google Meet

[Open event link](#)

Campus **10 May 2026, 12:00 am**

Women In Tech Career Panel

A conversation with alumni across Microsoft, Visa, HSBC, UBS, and Optum.

Hosted by Alumni Connect Team

Location: BVRITH Campus

Fig 9: Events Page

- An events page that showcases upcoming alumni meetups, workshops, and career sessions for students. It allows users to browse, filter, and access event details including date, mode, and participation links.
- The page helps students stay updated and actively engage in mentorship and career-building activities.

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Home About Directory Opportunities Events Forum Chat Dashboard [Open Portal](#)

Opportunities

Jobs, internships, and referral paths shared through the alumni network.

Career board

Students can browse opportunities and request referrals. Alumni can post opportunities from their own portal.

Open Roles

Browse current alumni-shared job and internship opportunities. 2 listed

Software Engineer Intern

Microsoft • Internship • Hyderabad OPEN

Work on product engineering, backend APIs, and platform reliability in a mentor-led summer internship.

Posted by Ashrita Guntur

[Apply Link](#) [Request Referral](#)

Associate Software Engineer

Infosys • Job • Bengaluru OPEN

Entry-level full-time opportunity for students graduating this year with strong fundamentals in Java, SQL, and problem solving.

Posted by G.Revathi Sai Nageswari

[Apply Link](#) [Request Referral](#)

Fig 10: Opportunities Page

- An opportunities page that lists jobs, internships, and referral openings shared by alumni. It allows students to explore roles, view details, and apply or request referrals easily.
- The page connects students with career opportunities and strengthens alumni-driven hiring support.

The implementation phase demonstrates a structured approach to developing a robust student-alumni mentorship and referral system. By leveraging modular design, this project ensures efficient processing, secure profile verification, and effective mentorship modeling. The integration of advanced evaluation metrics and a user-friendly deployment interface further emphasizes its practical applicability. The methodology outlined in this section not only achieves high performance but also lays the groundwork for future advancements, such as real-time messaging and integration with larger institutional datasets. This comprehensive approach showcases the potential of web-based systems in professional networking and career readiness portals.

6. CONCLUSION

In this project, we developed an advanced student-alumni mentorship and referral system using a modern web-based technology stack, which encompasses a wide range of professional networking activities. We meticulously prepared the data, verifying alumni profiles and formatting them for input into our relational database. Our system was built on the React and Spring Boot architecture, enhanced with role-based access control and secure messaging layers, allowing it to effectively process user inputs and manage professional interactions accurately. The platform development was carefully optimized using techniques such as manual admin verification and database checkpointing to ensure high performance and prevent data misuse. The evaluation of our system yielded promising results, with satisfactory user engagement and communication metrics on the validation set.

Visualizations of the system data flow and interaction matrix provided deeper insights into the platform's performance across different user roles, highlighting areas of strength and opportunities for further improvement. Additionally, our portal's ability to facilitate job referrals from industry professionals demonstrated its practical applicability, showcasing its potential for integration into real-world institutional systems. Overall, our project successfully illustrates the application of full-stack development to professional networking, achieving notable security and robustness. The challenges we encountered, such as data integrity and request handling, offered valuable learning experiences and insights into potential future enhancements. Moving forward, further research could explore more sophisticated automated verification, real-time video conferencing capabilities, and AI-driven mentorship matching. This project not only advances the field of campus networking but also sets a solid foundation for the development of intelligent systems that can support and enhance student career trajectories.

7. FUTURE SCOPE

The future scope of our Alumni Connect project is expansive and multi-dimensional, driven by rapid advancements in full-stack development, cloud computing, and database management. A key area for further exploration is improving the system's accuracy and efficiency through the integration of more sophisticated software architectures. For instance, adopting Microservices or more advanced versions of Spring Cloud could offer significant performance boosts. Additionally, leveraging machine learning techniques for automated mentor matching from datasets pre-trained on professional networking behavior would further enhance the platform's capabilities.

Expanding the dataset is another promising direction for improvement. Including a wider variety of institutional roles and utilizing synthetic data generation for testing referral algorithms could increase the system's robustness and versatility. Furthermore, transitioning towards real-time communication presents a fascinating challenge. Optimizing the portal for deployment on mobile applications and integrating it with real-time video conferencing frameworks would enable applications in domains that require immediate career guidance and response.

Another exciting avenue for exploration is the integration of multimodal data. By combining profile data with additional features such as professional portfolios, project repositories, and LinkedIn API integration, we can achieve a more nuanced and comprehensive understanding of student-alumni synergy. This multimodal approach can significantly enhance matching accuracy, especially in environments where basic profile data alone may be insufficient or ambiguous.

By continuously refining our system and exploring these emerging areas, we can push the boundaries of institutional networking, making it more accurate, efficient, and adaptable across various academic domains and professional applications.

8. REFERENCES

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APPENDIX

• Code Implementation

This section outlines the workflow of the Alumni Connect portal, including key code snippets for each functional step. The workflow is divided into several stages: database schema design, backend business logic, frontend interface development, and security implementation.

1. Frontend Dashboard Component

This code from the `dashboard` folder handles the User Interface, fetching data from the API and displaying it using Tailwind CSS.

File: `app/dashboard/page.tsx`

```
import { redirect } from "next/navigation";

import { DashboardOverview } from "@components/dashboard-overview";
import { getCurrentProfile, getSafeAuth } from "@lib/auth";
import { isClerkConfigured, missingAuthMessage } from "@lib/auth-config";

export default async function DashboardPage() {
  const { userId } = await getSafeAuth();

  if (!isClerkConfigured) {
    return (
      <>
      <section className="page-band">
        <div className="shell page-band-inner">
          <h1 className="page-title">Dashboard</h1>
          <p className="page-lead">Authentication is required to personalize the
alumni portal.</p>
        </div>
      </section>
      <section className="page-section">
        <div className="shell">
          <div className="empty-state">
            <strong>Authentication is not live yet.</strong>
            <p className="muted">{missingAuthMessage}</p>
          </div>
        </div>
      </section>
      </>
    );
  }

  if (!userId) {
    return <DashboardOverview />;
  }
}
```



```

let profile = null;

try {
  profile = await getCurrentProfile();
} catch {
  profile = null;
}

if (!profile) {
  return <DashboardOverview />;
}

if (profile.role === "ALUMNI") {
  redirect("/alumni/dashboard");
}

redirect("/student/dashboard");
}

```

2. Mentorship API Route

The "Controllers" are found in the `app/api/` folder. This code manages the backend logic for submitting mentorship applications.

File: `app/api/mentorship/route.ts`

```

import { NextRequest, NextResponse } from "next/server";

import { auth, clerkClient } from "@clerk/nextjs/server";

import { getSignedInIdentity } from "@lib/auth";
import { sendMentorshipStatusEmail } from "@lib/email";
import { prisma } from "@lib/db";
import { readPortalMentorshipRequests, updatePortalMentorshipRequest } from
"@lib/portal-store";

export async function POST(
  request: NextRequest,
  { params }: { params: Promise<{ id: string }> },
) {
  const { userId } = await auth();
  const { email } = await getSignedInIdentity();

  if (!userId) {
    return NextResponse.redirect(new URL("/sign-in", request.url));
  }

  let alumni = null;

```

```

try {
  alumni = await prisma.userProfile.findUnique({
    where: { clerkUserId: userId },
  });
} catch {
  alumni = null;
}

if (!alumni && email) {
  try {
    alumni = await prisma.userProfile.findUnique({
      where: { email },
    });
  } catch {
    alumni = null;
  }

  if (alumni && !alumni.clerkUserId) {
    try {
      alumni = await prisma.userProfile.update({
        where: { id: alumni.id },
        data: { clerkUserId: userId },
      });
    } catch {
    }
  }
}

if (!alumni || alumni.role !== "ALUMNI") {
  return NextResponse.redirect(new URL("/dashboard", request.url));
}

const formData = await request.formData();
const status = formData.get("status")?.toString() === "ACCEPTED" ? "ACCEPTED" :
"REJECTED";
const responseMessage = formData.get("responseMessage")?.toString().trim();
const { id } = await params;

let mentorshipRequest = null;

try {
  mentorshipRequest = await prisma.mentorshipRequest.findUnique({
    where: { id },
    include: { student: true, alumni: true },
  });
} catch {
  mentorshipRequest = null;
}

```

3. Global Layout and Styling

This section defines the structural wrapper of the application and the global styles, ensuring a consistent UI/UX across all roles.

File: app/layout.tsx

```
import type { Metadata } from "next";

import { ClerkProvider } from "@clerk/nextjs";

import { SiteFooter } from "@components/site-footer";
import { SiteHeader } from "@components/site-header";
import { isClerkConfigured } from "@lib/auth-config";

import "./globals.css";

export const metadata: Metadata = {
  title: "Alumni Connect",
  description:
    "A student-alumni mentorship, referral, and placement support portal for BVRIT Hyderabad.",
};

export default function RootLayout({
  children,
}: Readonly<{
  children: React.ReactNode;
}>) {
  const content = (
    <html lang="en">
      <body>
        <div className="page-chrome" />
        <SiteHeader />
        <main>{children}</main>
        <SiteFooter />
      </body>
    </html>
  );

  return isClerkConfigured ? <ClerkProvider>{content}</ClerkProvider> : content;
}
```

4. Email Service Configuration (Resend Integration)

This section defines the core email utility that interfaces with the Resend API. It ensures that the system can communicate with users asynchronously for critical status updates.

File: lib/mail.ts

```
import { Resend } from "resend";

type EmailPayload = {
  to: string;
  subject: string;
  html: string;
};

const resend = process.env.RESEND_API_KEY
  ? new Resend(process.env.RESEND_API_KEY)
  : null;

async function sendEmail({ to, subject, html }: EmailPayload) {
  if (!resend || !process.env.EMAIL_FROM) {
    return { skipped: true };
  }

  return resend.emails.send({
    from: process.env.EMAIL_FROM,
    to,
    subject,
    html,
  });
}

export async function sendMentorshipRequestEmail(input: {
  alumniEmail: string;
  alumniName: string;
  studentName: string;
  subject: string;
  goals: string;
  message: string;
}) {
  return sendEmail({
    to: input.alumniEmail,
    subject: `New mentorship request from ${input.studentName}`,
    html: `
      <div style="font-family: Arial, sans-serif; max-width: 640px; margin: 0 auto; color: #1f2937;">
        <h2>New mentorship request</h2>
        <p>Hello ${input.alumniName},</p>
        <p>${input.studentName} has requested mentorship through the BVRITH Alumni Portal.</p>
      `
  });
}
```

```

<p><strong>Topic:</strong> ${input.subject}</p>
  <p><strong>Goals:</strong> ${input.goals}</p>
  <p><strong>Message:</strong><br />${input.message}</p>
  <p>Please sign in to your dashboard to accept or reject the request.</p>
</div>
`,
});
}

export async function sendMentorshipStatusEmail(input: {
  studentEmail: string;
  studentName: string;
  alumniName: string;
  subject: string;
  status: "ACCEPTED" | "REJECTED";
  responseMessage?: string | null;
}) {
  const statusText = input.status === "ACCEPTED" ? "accepted" : "updated";

  return sendEmail({
    to: input.studentEmail,
    subject: `Your mentorship request was ${statusText}`,
    html: `
      <div style="font-family: Arial, sans-serif; max-width: 640px; margin: 0 auto;
color: #1f2937;">
        <h2>Mentorship request ${input.status.toLowerCase()}</h2>
        <p>Hello ${input.studentName},</p>
        <p>${input.alumniName} has ${input.status.toLowerCase()} your request about
<strong>${input.subject}</strong>.</p>
        ${
          input.responseMessage
            ? `<p><strong>Response from mentor:</strong><br
/>${input.responseMessage}</p>`
            : ""
        }
        <p>Please sign in to the portal to view the latest details.</p>
      </div>
    `
  });
  async function sendEmail({ to, subject, html }: EmailPayload) {
    if (!resend || !process.env.EMAIL_FROM) {
      return { skipped: true };
    }
    return resend.emails.send({
      from: process.env.EMAIL_FROM,
      to,
      subject,
      html,
    });
  }
}

```

```

export async function sendMentorshipRequestEmail(input: {
  alumniEmail: string;
  alumniName: string;
  studentName: string;
  subject: string;
  goals: string;
  message: string;
}) {
  return sendEmail({
    to: input.alumniEmail,
    subject: `New mentorship request from ${input.studentName}`,
    html: `
      <div style="font-family: Arial, sans-serif; max-width: 640px; margin: 0 auto;
color: #1f2937;">
        <h2>New mentorship request</h2>
        <p>Hello ${input.alumniName},</p>
        <p>${input.studentName} has requested mentorship through the BVRITH Alumni
Portal.</p>
        <p><strong>Topic:</strong> ${input.subject}</p>
        <p><strong>Goals:</strong> ${input.goals}</p>
        <p><strong>Message:</strong><br />${input.message}</p>
        <p>Please sign in to your dashboard to accept or reject the request.</p>
      </div>
    `,
  });
}

```